Nicholas J. Bryan, Ph.D. **Senior Research Scientist & Manager**

Email: njb@ieee.org Web: https://njb.github.io

SUMMARY

- AI researcher & manager w/12+ years' experience audio R&D (2+ years as manager)
- 11+ tech transfers, 39+ papers, 14+ patents, 9+ patents pending, IEEE Senior member
- Interested in generative models (diffusion, GANs, LLMs), content authenticity

EDUCATION

Ph.D., Computer-based Music Theory and Acoustics, Stanford University.
M.S., Electrical Engineering, Stanford University.
M.A., Music, Science, and Technology, Stanford University.
B.S., Electrical Engineering., U. of Miami-FL, Summa cum laude.
B.M., Music Eng. Tech., U. of Miami-FL, Summa cum laude.

WORK EXPERIENCE

- **Adobe Research** (San Francisco, CA)…..07/18-Now
  - Head of Music AI Group (12/23 – Now)
    - Led 0-to-1 cross-org effort to ship generative music at Adobe
    - Shipped industry-first studio-quality, commercially safe GenAI music for *any* use.
    - Owner of all modeling and data work for Firefly Audio Model
  - Senior Research Scientist II (02/22 – Now)
  - Senior Research Scientist I (07/20 – 02/22)
  - Research Scientist II (07/18 – 07/20)
- **Apple Inc.**, Interactive Media Group (Cupertino, CA)...................................01/14-07/18
  - Senior Audio Algorithm Engineer (10/16 – 07/18)
  - Audio Algorithm Engineer (01/14 – 10/16)

TECHNOLOGY TRANSFERS

- Adobe Technologies
  - Beats (lead) – Adobe Premiere Mobile, 2026.
  - Audio Match (lead) – Adobe Stock AI Studio, 2026.
  - Generate Soundtrack/Firefly Audio Model (lead) – Adobe Firefly (web & mobile), 2025.
  - Filler Word Removal (consult) – Premiere Pro, 2023.
  - Convertible Melspectrograms (lead) – Premiere Pro, 2023.
  - Voice Activity Detection (consult) – Premiere Pro, 2023.
  - AutoTone Lumetri Color (consult) – Premiere Pro, 2022.

- AudioTagger (co-lead research) – [Adobe Project Blink](), 2022.
- MicCheck (lead) – [Adobe Podcast](), 2022.
- Find Similar Audio (co-lead research, lead engineering) – [Adobe Stock](), 2021.
- Sensei Toolkit (lead) – [Sensei Machine Learning Platform](), 2020.
- Merge Clips (lead) – [Premiere Pro](), 2013.
- PR Events: Three MAX Sneaks + 1 PR Event

- Apple – Single & multi-channel speech enhancement for iPhone + AirPods Pro 2014-2018.

JOURNAL PUBLICATIONS (PEER-REVIEWED)

7. Y. Bai, J. Casebeer, S. Sojoudi, N. J. Bryan, "DRAGON: Distributional Rewards Optimized Diffusion Generative Models." Transactions on Machine Learning Research (TMLR), 2025.

6. G. Zhu, J.P. Caceres, Z. Duan, N. J. Bryan, "MusicHiFi: Fast High-Fidelity Stereo Vocoding." *IEEE Signal Processing Letters (SPL),* 2024.

5. S-L. Wu, C. Donahue, S. Watanabe, and N. J. Bryan, "Music ControlNet: Multiple Time-varying Controls for Music Generation." IEEE Trans. on Audio, Speech, and Lang. Proc. (TASLP), 2023.

4. J. Casebeer, N. J. Bryan, P. Smaragdis, "MetaAF: Meta-learning for Adaptive Filtering." IEEE Transactions on Audio, Speech, and Language Processing (TASLP), 2023.

3. C. J. Steinmetz, N. J. Bryan, J. D. Reiss. "Style Transfer of Audio Effects with Differentiable Signal Processing." Journal of Audio Engineering Society (JAES), 2022.

2. Z. Tang, N. J. Bryan, D. Li, T. Langlois, D. Manocha, "Scene-Aware Audio Rendering via Deep Acoustic Analysis." IEEE VR Journal Papers, 2020.

1. T. Quirino, M. Kubat, N. J. Bryan. "Instinct-Based Mating in Genetic Algorithms Applied to the Tuning of 1-NN Classifiers." *IEEE Trans. on Knowledge and Data Engineering.* December 2010.

CONFERENCE PUBLICATIONS (PEER-REVIEWED)

37. [Under Review] Y-B. Lin, J. Casebeer, L. Mai, A. Mahapatra, G. Bertasius, N. J. Bryan. "V2M-Zero: Zero-Shot Time-Aligned Video-to-Music Generation." Under Review, 2025.

36. I. Bukey, Z. Wang, C. Donahue, N. J. Bryan. "Rethinking Music Captioning with Music Metadata LLMs." *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2026.

35. J. Casebeer, G. Zhu, Zhepei Wang, N. J. Bryan. "A Generative-first Neural Audio Autoencoder." *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2026.

34. S-L. Wu, G. Zhu, J-P. Caceres, C-Z. Huang, N. J. Bryan. "Stemphonic: All-at-once Flexible Multi-stem Music Generation." *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2026.

33. Z. Novack, G. Zhu, J. Casebeer, J. McAulay, T. Berg-Kirkpatrick, N. J. Bryan, "Presto! Distilling Steps and Layers for Accelerating Music Generation*." International Conference on Learning Representations (ICLR)* ***(Spotlight, top 5%)***, 2025.

32. Z. Novack, J. McAulay, T. Berg-Kirkpatrick, N. J. Bryan, "DITTO-2: Distilled Diffusion Inference Time T-Optimization for Music Generation." *International Society of Music Information Retrieval (ISMIR),* 2024.

31. Z. Novack, J. McAulay, T. Berg-Kirkpatrick, N. J. Bryan, "DITTO: Diffusion Inference-Time T-Optimization for Music Generation*." International Conference on Machine Learning (ICML)* ***(Oral, top 1.5%)****,* 2024.

30. J. Wu, J. Casebeer, N. J. Bryan, P. Smaragdis, "Meta-learning for Adaptive Filters with Higher-order Frequency Dependencies." *IEEE International Workshop on Acoustic Signal Enhancement (IWEANC)*, 2022.

29. H. Yang, S. Firodiya, N. J. Bryan, M. Kim. "Don't Separate, Learn to Remix: End-to-End Neural Remixing with Joint Optimization," *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2022.

28. M. Won, J. Salamon, N. J. Bryan, G. J. Mysore, X. Serra, "Emotion Embedding Spaces for Matching Music to Stories." *International Society for Music Information Retrieval Conference (ISMIR)*. Online, 2021. **(Best student paper award)**

27. J. Salamon, O. Nieto, N. J. Bryan. "Deep Embeddings and Section Fusion Improve Music Segmentation." *International Society for Music Information Retrieval Conference (ISMIR)*, 2021.

26. Y. Wang, N. J. Bryan, J. Salamon, M. Cartwright, J. P. Bello. "Who Calls the Shots? Rethinking Few-Shot Learning for Audio." *IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA)*, 2021. **(IEEE special best paper award).**

25. J. Casebeer, N. J. Bryan, P. Smaragdis. "Auto-DSP: Learning to Optimize Acoustic Echo Cancellers." *IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA)*, 2021.

24. M. A. Martínez Ramírez, O. Wang, P. Smaragdis, N. J. Bryan. "Differentiable Signal Processing with Black-Box Audio Effects." *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP),* 2021.

23. Y. Wang, N. J. Bryan, M. Cartwright, J. P. Bello, J. Salamon. "Few-shot Continual Learning for Audio Classification." *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP),* 2021.

22. M. Morrison, L. Rencker, Z. Jin, N. J. Bryan, J.-P. Caceres, B. Pardo. "Context-aware Prosody Correction for Text-based Speech Editing." *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP),* 2021.

21. P. Manocha, A. Finkelstein, Z. Jin, R. Zhang, N. J. Bryan, G. J. Mysore. "A Differentiable Perceptual Audio Metric Learned from Just Noticeable Differences." *Interspeech*, 2020. **(Best student paper finalist)**

20. M. Morrison, Z. Jin, J. Salamon, N. J. Bryan, G. J. Mysore. "Controllable Neural Prosody Synthesis." *Interspeech*, 2020.

19. J. Lee, N. J. Bryan, J. Salamon, Z. Jin, J. Nam. "Metric Learning vs Classification for Disentangled Music Representation Learning." *International Society for Music Information Retrieval (ISMIR),* 2020.

18. Y. Wang, J. Salamon, M. Cartwright, N. J. Bryan, J. P. Bello. "Few-shot Drum Transcription in Polyphonic Music." *International Society for Music Information Retrieval (ISMIR),* 2020.

17. J. Lee, N. J. Bryan, J. Salamon, Z. Jin, J. Nam. "Disentangled Multidimensional Metric Learning for Music Similarity." *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP),* 2020.

16. Y. Wang, J. Salamon, N. J. Bryan, J.-P. Bello. "Few-Shot Sound Event Detection." *IEEE Internation Conference on Acoustics, Speech, and Signal Processing (ICASSP),* 2020. **(Press)**

15. S. I. Mimilakis, N. J. Bryan, P. Smaragdis. “One-Shot Parametric Audio Production Style Transfer with Application to Frequency Equalization.” *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP),* 2020.

14. N. J. Bryan. “Impulse Response Data Augmentation and Deep Neural Networks for Blind Room Acoustic Parameter Estimation.” *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP),* 2020.

13. N. J. Bryan, G. J. Mysore, G. Wang. “ISSE: An Interactive Source Separation Editor.” *Conference on Human Factors in Computing Systems (CHI),* 2014.

12. N. J. Bryan. “Interactive Sound Source Separation.” *Ph.D. Thesis*, Stanford University, 2014. **Audio Eng. Society Graduate Student Design Gold Award (2013).**

11. N. J. Bryan, G. J. Mysore, G. Wang. “Source Separation of Polyphonic Music with Interactive User-Feedback on a Piano-Roll Display.” *International Society for Music Information Retrieval Conference (ISMIR),* 2013.

10. N. J. Bryan, G. J. Mysore. “An Efficient Posterior Regularized Latent Variable Model for Interactive Sound Source Separation.” *International Conference on Machine Learning (ICML),* 2013. **(Oral)**

9. N. J. Bryan, G. J. Mysore. “Interactive Refinement of Supervised and Semi-Supervised Sound Source Separation Estimates.” *IEEE Int. Conference on Acoustics, Speech, and Signal Processing (ICASSP),* 2013.

8. N. J. Bryan, J. Herrera, G. Wang. “User-Guided Variable-Rate Time-Stretching Via Stiffness Control.” *International Conference on Digital Audio Effects (DAFx)*, 2012.

7. N. J. Bryan, P. Smaragdis, G. J. Mysore. “Clustering and Synchronizing Multi-Camera Video via Landmark Cross-Correlation.” *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP),* 2012.

6. N. J. Bryan, G. Wang. “Musical Influence Network Analysis and Rank of Sample-Based Music.” *International Society for Music Information Retrieval Conference (ISMIR),* 2011.

5. N. J. Bryan, G. Wang. “Two Turntables and a Mobile Phone.” *International Conference on New Interfaces for Musical Expression (NIME),* 2011. **(Press)**

4. K. Lee, N. J. Bryan, J. S. Abel. “Approximating Measured Reverberation Using a Hybrid Fixed/Switched Convolution Structure.” *International Conference on Digital Audio Effects DAFx),* 2010.

3. N. J. Bryan, J. Herrera, J. Oh, G. Wang. “MoMu: A Mobile Music Toolkit.” *International Conf. on New Interfaces for Musical Expression (NIME)*, 2010. **(Press)**

2. J. Oh, J. Herrera, N. J. Bryan, G. Wang. “Evolving The Mobile Phone Orchestra.” *International Conference on New Interfaces for Musical Expression (NIME)*, 2010.

1. G. Wang, N. Bryan, J. Oh, R. Hamilton. “Stanford Laptop Orchestra (SLORK).” *International Computer Music Conference (ICMC)*, 2009. **(Press)**

OTHER PUBLICATIONS

7. J. Casebeer, N. J. Bryan, P. Smaragdis, “Scaling Up Adaptive Filter Optimizers.” *arXiv,* 2024.

6. M. Morrison, Z. Jin, N. J. Bryan, J.-P. Caceres, B. Pardo, “Neural Pitch-Shifting and Time-Stretching with Controllable LPCNET.” arXiv*, 2021.*

5. N. J. Bryan, G. J. Mysore. “Interactive User-Feedback for Sound Source Separation.” *International Conference on Intelligent User-Interfaces (IUI) Workshop on Interactive Machine Learning (Extended Abstract/Workshop Paper).* March 2013.

4. J. S. Abel, N. J. Bryan, T. Skare, M. Kolar, P. Huang, D. Mostowfi, J. O. Smith III. “A Configurable Microphone Array with Acoustically Transparent Omnidirectional Elements.” *In Audio Engineering Society Convention.* October 2009.

3. N. J. Bryan, M. A. Kolar, J. S. Abel. “Impulse Response Measurements in the Presence of Clock Drift.” In *Audio Engineering Society Convention.* November 2010.

2. N. J. Bryan, J. S. Abel. “Methods For Extending Room Impulse Responses Beyond Their Noise Floor.” In *Audio Engineering Society Convention.* November 2010.

1. J. S. Abel, N. J. Bryan, et al. “On Estimating Room Impulse Responses from Recorded Balloon Pops.” In *Audio Eng. Society Convention.* November 2010.

AWARDS

- Adobe Impact Award, 2025.
- IEEE Senior Member, 2024.
- Adobe Distinguished Inventor, 2023.
- Adobe Intern Showcase 1st Prize (Mentor), 2023.
- Best Student Paper Award (mentor), ISMIR, 2021.
- IEEE Best Audio Few-shot Learning Paper, IEEE WASPAA, 2021.
- Best Student Paper Finalist, Interspeech, 2020.
- Best Reviewer Award, ISMIR, 2020.
- Gold Award, AES Graduate Student Design, 2013.
- U of Miami-FL, Summa cum laude honors, General honors, Dept. of EE honors 2007.
- U of Miami-FL, Tzay Y. Young Best EE Senior Design Award, U of Miami-FL, 2007.
- Frost Wind Ensemble Performance, Carnegie Hall, 2007.
- U of Miami-FL, Research & Creativity Fair, Undergrad 2nd Place Winner, 2006.

OPEN-SOURCE SOFTWARE

- **Convmelspec** – Convertible Melspectrograms via 1D Convolutions (https://github.com/adobe-research/convmelspec), released 2022.
- **MetaAF** – Control adaptive filters with neural networks (https://github.com/adobe-research/MetaAF), released 2022.
- **DeepAFx-ST** – Style transfer of audio effects with differentiable signal processing (https://github.com/adobe-research/DeepAFx-ST), released 2022.
- **DeepAFx –** Third-party audio effects plugins as differentiable layers within deep neural networks (https://github.com/adobe-research/DeepAFx), released 2021.
- **ISSE** – Cross-platform interactive source separation editor via drawing and painting

(http://isse.sourceforge.net). 82,000+ downloads, 120+ countries. released 2013.

PATENTS SUBMITTED

- Z. Novack, G. Zhu, J. Casebeer, N. J. Bryan, "Accelerated Diffusion Transformer Audio Generation, 2025.
- G. Zhu, J.P. Caceres, N. J. Bryan. "MusicHiFi: An Efficient High-Fidelity Vocoder", 2025.
- J. Casebeer, N. J. Bryan, "Meta-Learning for Adaptive Filters", 2025.
- S-L. Wu and N. J. Bryan, "Music Generation with Time Varying Controls", 2024.
- C. J. Steinmetz, N. J. Bryan, Simon Jenni, and John Collomosse, "General Purpose Audio Fingerprinting", 2023
- J. Salamon, J.-P. Caceres, G. Zhu, N. J. Bryan**.** "A Pipeline for Filler Word Detection and Classification." Submitted by Adobe, August 2022.
- N. J. Bryan, P. Smaragdis. "Method for Learning to Optimize Signal Processing Algorithms." Submitted by Adobe, April 2022.
- J. Salamon, O. Nieto, N. J. Bryan, "Multi-Level Audio Segmentation Using Deep Embeddings." Submitted by Adobe, Oct. 2022.
- N. J. Bryan and J. Salamon, "Section-Based Music Similarity Searching." Submitted by Adobe, Oct. 2022.

PATENTS ISSUED

- M. Martínez, N. J. Bryan, O. Wang, P. Smaragdis, "Deep Encoder for Performing Audio Processing", Adobe, US 11,900,902, 2024.
- M. Morrison, Z. Jin, J. P. Caceres, N. J. Bryan, B. Pardo. "Neural Pitch-Shifting and Time-Stretching" Adobe, US 11,915,714, 2024.
- M. Morrison, Lucas Rencker, Z. Jin, N. J. Bryan, Juan-Pablo Caceres, Bryan Pardo. "Context-Aware Prosody Correction of Edited Speech", Adobe, US 11,830,48, 2023.
- J. Lee, N. J. Bryan, J. Salamon, Z. Jin, "Searching for Music." Adobe, US 11,636,342 March 2020.
- N. J. Bryan. "Improving Voice Recordings Using Acoustic Quality Measurement Models and Actionable Acoustic Improvement Suggestions." Adobe, US 11,462,236, 2019.
- J. Salamon, Y. Wang, N. J. Bryan, "Automated Sound Matching Within an Audio Recording." Adobe, US 11,501,102,2019.
- N. J. Bryan, Q. Yang, V. Iyengar, "Mitigating Noise in Audio Signals". Apple, 2021.
- S. I. Mimilakis, N. J. Bryan, P. Smaragdis, "Audio Prod. Assistant for Style Transfers of Audio Recordings Using One-Shot Parametric Predictions", Adobe, US 11,082,789 2021.
- D. Li, Z. Tang, T. Langlois, N. J. Bryan. "Rendering Scene-Aware Audio Using Neural Network-Based Acoustic Analysis." Adobe, US 11,190,898, 2021.
- N. J. Bryan. "Generating Synthetic Acoustic Impulse Responses from an Acoustic Impulse Response." US11,074,925, Adobe, 2021.
- N. J. Bryan, V. Iyengar, "Speech Enhancement for an Electronic Device." US20190272842, Apple, 2020.

- N. J. Bryan, V. Iyengar, A. Lindahl, “Transparent Near-End User Control Over Far-End Speech Enhancement Processing”, US20190156847A1, Apple, 2019.
- N. J. Bryan, V. Iyengar, “System and Method of Noise Reduction for Mobile Device.” US20180350381A1. Apple, 2019
- N. J. Bryan, P. Smaragdis, G. J. Mysore, “Clustering and Synchronizing Content.” US8924345B2, Adobe, 2014.

PhD THESIS COMMITTEES

- Zachary Novack – UCSD, current.
- Ge Zhu – University of Rochester, 2025.
- Eloi Moliner Juanpere – Alto University (external examiner), 2025.
- Jonah Casebeer – University of Illinois Urbana-Champaign, 2023
- Yu Wang – New York University, 2023
- Jongpil Lee – Korea Advanced Institute for Science and Technology, 2020

TECHNICAL COMITTEES

- IEEE Audio and Acoustic Signal Processing (AASP) Technical Committee (2023-2025)
- IEEE Audio and Acoustic Signal Processing (AASP) Technical Committee (2020-2022)

AREA CHAIR, META REVIEWER, SESSION CHAIRS

- ISMIR 2026 Meta Reviewer
- ACM Creativity and Cognition Conference – Meta Reviewer (6)
- ICASSP 2026 AASP TC Area Chair – Audio Security (14)
- ISMIR 2025 Meta Reviewer (5)
- IEEE WASPAA 2025 Area Chair (5)
- IEEE ICASSP 2025 AASP TC Area Chair—Data & Open Source (16)
- ISMIR 2024 Meta-Reviewer (5)
- IEEE ICASSP 2024 AASP TC Area Co-Chair– Music Signal Analysis, Proc, and Synth (38)
- IEEE AASP TC – EDICS Vice-Chair 2025
- IEEE WASPAA 2023 General Co-Chair
    - Co-led review process transition from single- to double-blind
    - Co-led 2-3x increase in corporate sponsorship donations
    - Co-led 10x increase in student travel grants
    - Led first individual donation campaign w/IEEE Foundation
    - Co-led 2-2.5x increase in student attendance at WASPAA

- IEEE ICASSP 2023 Area Chair – Music Signal Analysis, Processing, and Synthesis (34)
- ISMIR 2022 Meta Reviewer (3)
- IEEE ICASSP 2022 Area Chair – Music Signal Analysis, Processing, and Synthesis (41)
- ISMIR 2021 Meta Reviewer (3)
- IEEE WASPAA 2021 Area Chair – Music Signal Analysis, Processing, and Synthesis (12)
- IEEE ICASSP 2021 AASP – Deep Learning for Music Signal Processing Session Chair
- IEEE ICASSP 2021 AASP – Music Signal Analysis, Processing & Synthesis Area Chair (24)
- IEEE ICASSP 2020 AASP – Music Information Retrieval Session Chair

ACADEMIC REVIEWING

- 2026 – ACM C&C (5), ECCV (1), NeurIPs (1)
- 2025 – WASPAA (5), ISMIR (5), TMLR (1)
- 2024 – TASLP (1), SIGGRAPH (1)
- 2023 – TASLP (3), SPL (1)
- 2022 – IEEE SPL (1), IEEE TASLP (3)
- 2021 – IEEE TASLP (2), ISMIR reviews (3), Pattern Recognition (1)
- 2020 – ICASSP (12), SIGGRAPH (1), IEEE TASLP (3), ISMIR (4, **Best Reviewer**)
- 2019 – ICASSP (10), WASPAA (6), IEEE TSP (1), IEEE SPL (1), Stanford HAI Blog (1)
- 2018 – ICASSP (3), EUSIPCO (5), NIME
- 2017 – ICASSP, WASPAA, EUSIPCO, ACM Trans. on Interactive Int. Systems, NIME
- 2016 – ICASSP, WASPAA, EUSIPCO, ACM Trans. on Interactive Int. Systems, NIME
- 2015 – ICASSP, WASPAA, EUSIPCO, JASA, NIME, IEEE Journal Topics Signal Proc.
- 2014 – ICASSP, Elsevier Signal Processing, NIPS, NIME, Computer Communications

SELECTED PRESS

- Generate Soundtrack, Adobe News + 2x Keynote Demos (10/28, 2025) https://news.adobe.com/news/2025/10/adobe-max-2025-firefly
- Project Music GenAI Control, Adobe News (2/2024) https://blog.adobe.com/en/publish/2024/02/28/adobe-research-audio-creation-editing
- MAX Sneaks – "SoundSeek." (11/19). https://www.youtube.com/watch?v=ebAlnW3Xs2k
- MAX Sneaks – "Audio Layers." Adobe TV (05/13). http://www.youtube.com/watch?feature=player_embedded&v=nyODK-J9keo
- MAX Sneaks – "Automatic Synchronization of Crowd Sourced Videos." Adobe TV (10/11). https://www.youtube.com/watch?v=dhyvzOJBhzc
- "Smartphone turntables put new spin on DJing." NewScientist *(*05/11). https://tinyurl.com/newscientist-mophodj
- "Air Instruments only." *The Stanford Daily* (10/10). http://www.stanforddaily.com/2010/10/28/air-instruments-only

- "From Pocket to Stage, Music in the Key of iPhone." NY Times *(Front Page* 12/04/09). http://www.nytimes.com/2009/12/05/technology/05orchestra.html?_r=2&hpw
- "Make-TV: Computer Making Music (Episode 9)." PBS (02/09). https://www.youtube.com/watch?v=R2OF2OLaHdw
- "Stanford Students invent new music." Local ABC TV (SF), Drive to Discover (04/08). http://abclocal.go.com/kgo/story?section=news/drive_to_discover&id=6064990
- "Rhythm and Cubes." Stanford Magazine (07/08). http://www.stanfordalumni.org/news/magazine/2008/julaug/red/cubeats.html

MENTORING

- Irmak Bukey (2025) – CMU
- Nithya Shikarpur (2025) – MIT
- Dimitrios Bralios (2025) – UIUC
- Yan-Bo Lin (2025) – UNC
- SeungHeon Doh (2024) – KAIST
- Yatong Bai (2024) – University of California Berkeley
- Zachary Novack (2023-2024) – University of California San Diego
- Ge Zhu (2023) – University of Rochester
- Shih-Lun Wu (2023, 2025) – CMU, MIT
- Jonah Casebeer (2020-2022) – University of Illinois Urbana-Champaign
- Christian Steinmetz (2021, 2022) – Queen Mary University of London
- Jack Atherton (2021) – Stanford University
- Gabriel Zalles (2021) – University of California San Diego
- Amber Blue (2021) – Georgia Tech
- Minz Won (2020-2021) – Universitat Pompeu Fabra
- Marco Antonio Martínez Ramírez (2020) – Queen Mary University of London
- Yu Wang (2019-2021) – New York University
- Jongpil Lee (2019-2020) – Korea Advanced Institute for Science and Technology
- Stylianos Mimilakis (2019) – Technical University of Ilmenau
- Lucas Rencker (2019) – University of Surrey
- Zhenyu Tang (2019) – University of Maryland
- Maxwell Morrison (2019-2021) – Northwestern University

SELECTED TALKS & WORKSHOPS

- "Training Music Foundation Models", Conversational AI Reading Group at Mila, May 2026.

- “Creative AI: Music, Authorship, and Algorithmic Culture,” UCSD Design@Large Speaker Series Invited Speaker (of three guests), April 2026.
- "Beyond Text to Music" Keynote Talk, GenDA Workshop, ICASSP 2025.
- “My Career Journey”, University of Miami Music Engineering Forum, 2025.
- “Learning to Control Signal Processing Algorithms with Deep Learning”, *Women in Music Information Retrieval Workshop*, 2021.
- “ISSE: An Interactive Machine Learning Workflow.” (Workshop Demo) NIPS *Workshop on Machine Learning Open-Source Software.* December 2013.
- “ISSE: An Interactive Source Separation Editor.” (Two-part Talk) *CCRMA DSP Seminar*, Stanford University. October and November 2013.
- “Audio Source Separation.” (Workshop Speaker w/Gautham J. Mysore, Pierre Leveau, Derry FitzGerald, Elias Kokkinis), *Audio Eng. Society Convention*. October 2013.
- “Source Separation Tutorial Mini-Series.” (Three-part Talk w/Dennis Sun and Eunjoon Cho) *CCRMA DSP Seminar*, Stanford University, April 2013.